UNIT 3: THERMOMETRY & FIBER OPTICS

Heat, temperature and temperature scales

Heat & temperature

Heat is energy transferred from a hotter body to a colder one. Temperature measures how hot or cold a body is and is related to the average kinetic energy of its particles.

Quantity	SI unit	Other units / note
Heat	joule (J)	1 cal ≈ 4.184 J; 1 kcal = 1000 cal
Temperature	kelvin (K)	degree Celsius (°C), degree Fahrenheit (°F)

- Heat is an extensive quantity: it depends on the amount of material. Temperature is intensive: it does not depend on sample size.

Heat is the cause; temperature change is an effect.

Temperature conversions

K = °C + 273.15 (often rounded to 273 in class problems)
°F = (9/5)°C + 32 • °C = (°F − 32) × 5/9

Example	Conversion	Answer
100°C	K = 100 + 273; °F = 1.8×100 + 32	≈ 373 K; 212°F
98.6°F	°C = (98.6−32)×5/9; K ≈ °C + 273	37°C; ≈ 310 K
300 K	°C ≈ 300−273; °F = 1.8×27 + 32	27°C; 80.6°F

The Kelvin scale begins at absolute zero (0 K ≈ −273.15°C). Water freezes at 0°C / 32°F and boils at 100°C / 212°F at standard atmospheric pressure.

HEAT TRANSFER

Conduction, convection and radiation

Heat flows spontaneously from a higher-temperature region to a lower-temperature region. There are three main modes:

Mode	How heat moves	Where it occurs / example
Conduction	Energy passes between particles without bulk movement	Mainly solids; a metal spoon warms in hot tea
Convection	Heat moves with the actual motion of fluid particles	Liquids and gases; warm air rises
Radiation	Energy travels as electromagnetic waves; no material medium needed	Sunlight warming Earth

Conductors and insulators of heat

- Good conductors: heat passes through readily; examples include copper, iron, silver, aluminium and steel.
- Insulators (bad conductors): slow heat transfer; examples include wood, plastic and glass.

Thermal conductivity

For steady conduction through a uniform rod, heat transferred Q increases with area A, temperature difference ($\theta_1 - \theta_2$), and time t; it decreases as length d increases.

$$Q = k\,A\,(\theta_1 - \theta_2)\,t / d$$

k is the coefficient of thermal conductivity. SI unit: W/(m·K), equivalent to J/(m·K·s).

GAS LAWS & THE IDEAL GAS EQUATION

For a fixed amount of gas

Boyle's law — constant temperature

Pressure varies inversely with volume: when volume decreases, pressure increases.

$P \propto 1/V$ • $PV = \text{constant}$ • $P_1V_1 = P_2V_2$

Charles's law — constant pressure

Volume is directly proportional to absolute temperature.

$V \propto T$ • $V/T = \text{constant}$ • $V_1/T_1 = V_2/T_2$

Gay-Lussac's law — constant volume

Pressure is directly proportional to absolute temperature.

$P \propto T$ • $P/T = \text{constant}$ • $P_1/T_1 = P_2/T_2$

Use kelvin for gas-law temperatures: $T(K) = t(°C) + 273.15$.

Ideal gas equation

$$PV = nRT$$

P = pressure, V = volume, n = amount in moles, T = absolute temperature, and R = universal gas constant ($8.314\ J \cdot mol^{-1} \cdot K^{-1}$).

- These relationships describe how pressure, volume and temperature change together for a fixed amount of gas.

SPECIFIC HEAT CAPACITY

How much heat raises a substance's temperature?

Specific heat capacity, c

Specific heat capacity is the heat required to raise the temperature of 1 kg of a substance by 1 K (or 1°C). Different materials warm by different amounts for the same heat input.

$$Q = m\,c\,\Delta T \bullet c = Q / (m\,\Delta T)$$

Q = heat transferred; m = mass; ΔT = temperature change. SI unit of c: J/(kg·K).

Specific heats of gases

Symbol	Condition	Meaning
Cp	Constant pressure	Heat capacity per unit mass (or per mole) at fixed pressure
Cv	Constant volume	Heat capacity per unit mass (or per mole) at fixed volume

- For a gas, Cp is greater than Cv: at constant pressure, some supplied energy also does expansion work.

For molar heat capacities: $C_p - C_v = R$.
Ratio of specific heats: $\gamma = C_p/C_v$.

The source notes use c for specific heat and Q for heat. Keep mass and units consistent when substituting into $Q = mc\Delta T$.

REFLECTION, REFRACTION & TOTAL INTERNAL REFLECTION

Light changes direction at surfaces and boundaries

Reflection

Reflection occurs when light strikes a surface and returns in the same medium.

> **Law of reflection: angle of incidence i = angle of reflection r.**
> **Incident ray, reflected ray and the normal lie in the same plane.**

Refraction

Refraction is the change in direction of light as it enters a different transparent medium, caused by a change in speed.

- From a rarer to a denser medium, a ray bends toward the normal; from denser to rarer, it bends away.

> **Snell's law: $n_1 \sin i = n_2 \sin r$**
> **Refractive index (absolute): $n = c/v$**

Total internal reflection (TIR)

When light travels from a denser medium to a rarer medium, the angle of refraction grows with the angle of incidence. At the critical angle ic, the refracted ray is at 90°; for i > ic, all light reflects back into the denser medium.

> **TIR conditions: (1) denser to rarer medium; (2) incidence angle greater than the critical angle.**
> **For $n_1 > n_2$: $\sin ic = n_2/n_1$.**

OPTICAL FIBERS

Structure, working principle and applications

What is an optical fiber?

A very thin transparent fiber carries light signals over distance with little loss. It works by repeated total internal reflection.

Part	Role
Core	Inner glass or plastic path; higher refractive index
Cladding	Surrounds the core; lower refractive index keeps light inside
Jacket	Outer protective covering

How it guides light

Light enters the core within an allowed range. At the core–cladding boundary it meets the TIR condition, reflects repeatedly, and travels along the fiber.

Acceptance angle θa: largest entry angle (measured from the fiber axis) that still lets light propagate.
For air outside: NA = sin θa = √(ncore² − ncladding²).

Fiber types

Classification	Types	Typical use
Refractive index	Step-index; graded-index	Graded index changes gradually from core axis outward
Light propagation	Single-mode; multimode	Single-mode: narrow core, laser, long distance. Multimode: wider core, LED, shorter distance.
Material	Glass; plastic	Chosen for the application and required transmission

Applications

- High-speed internet and computer networks; cable television.
- Medicine and research, including endoscopy; mechanical inspection.
- Automotive lighting, decorative lighting and communications.